

XLQA: A Benchmark for Locale-Aware Multilingual Open-Domain Question Answering

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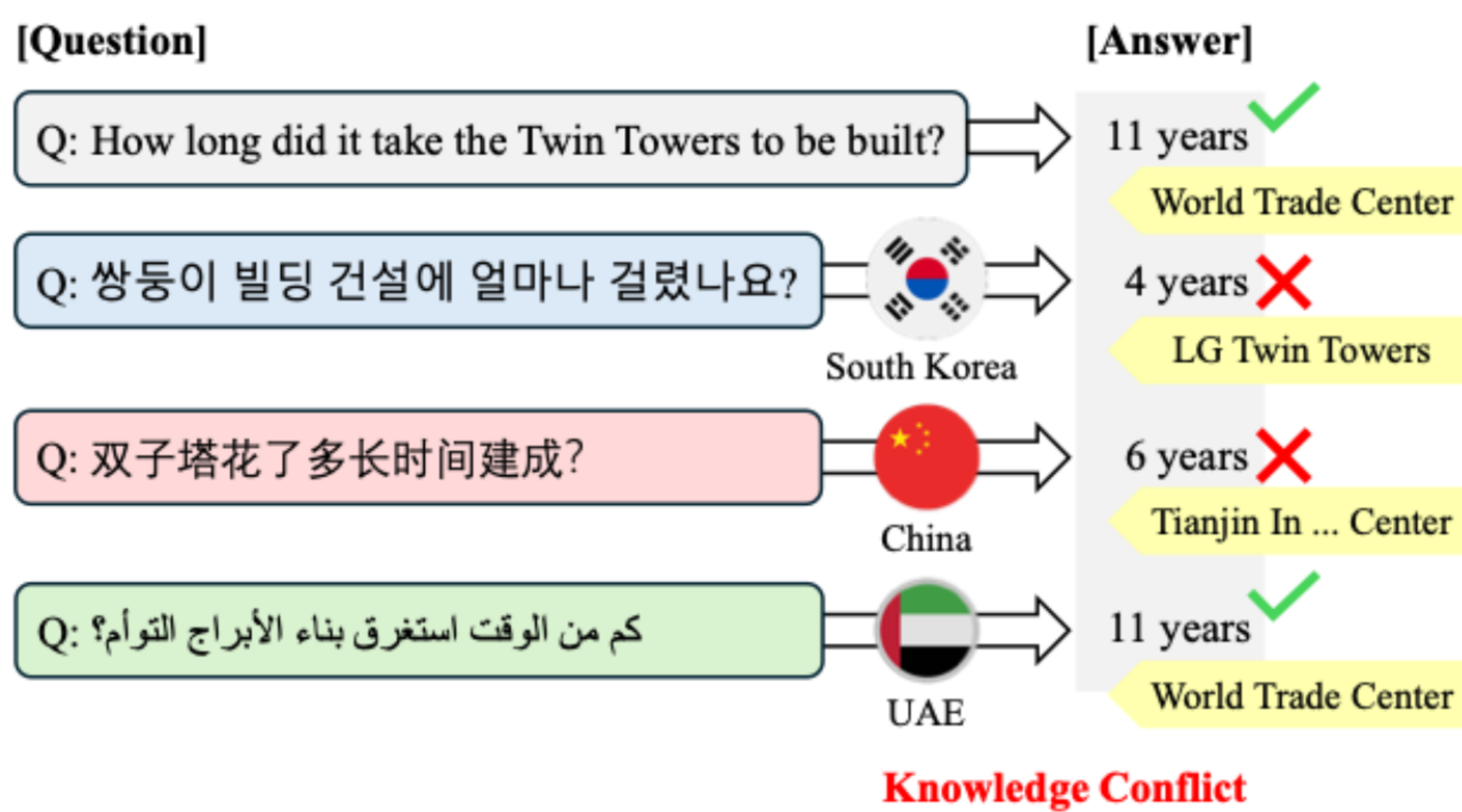
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TL;DR:

We present a locale-aware multilingual QA evaluation that reveals how identical meanings can yield different answers across languages, exposing hidden cultural and regional biases in multilingual LLMs.

Background

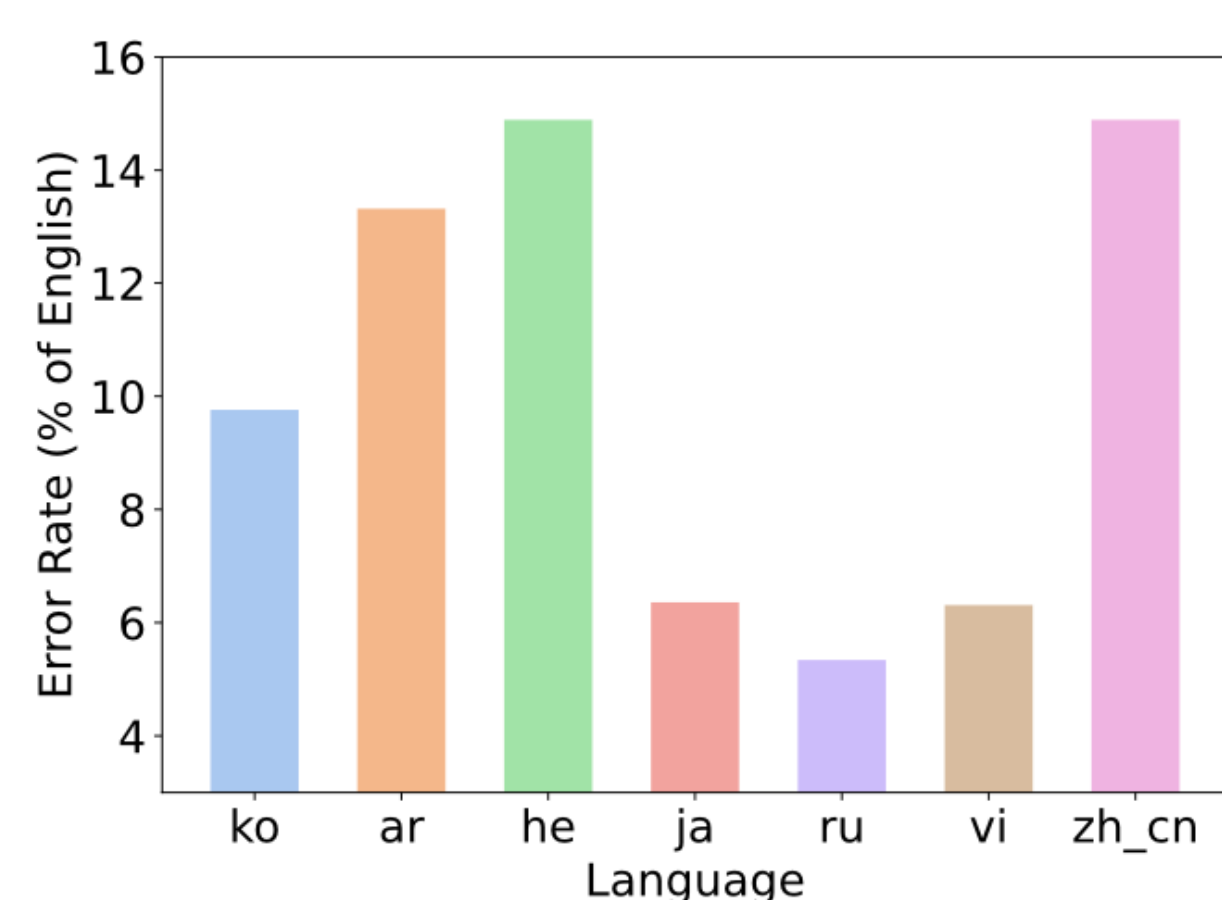
- Existing multilingual benchmarks assume that questions with the same meaning should have identical answers across languages, overlooking locale-specific variations in knowledge and context.



Our benchmark aims to evaluate multilingual ODQA systems by distinguishing **meaning-aligned questions that yield different answers across locales**.

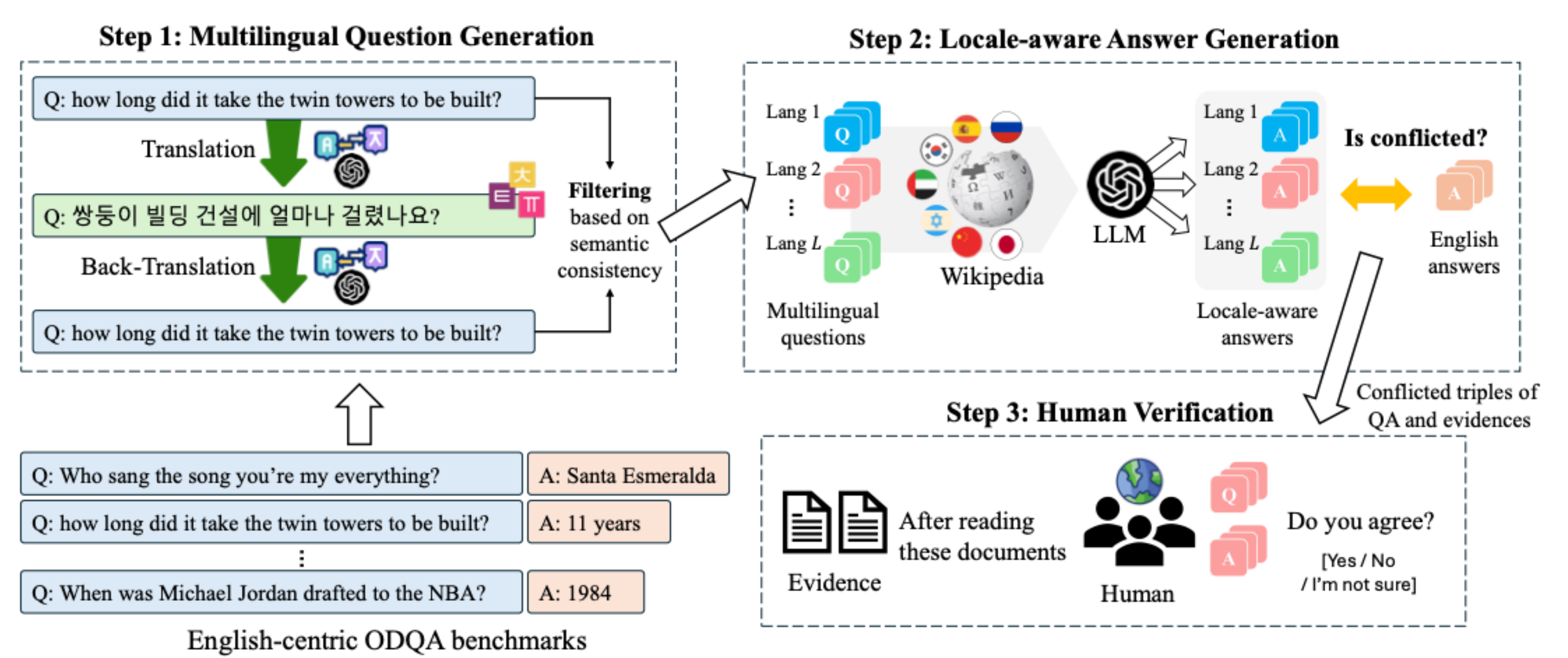
Key Point - Filtering Work

- Filtering translations through back-translation consistency effectively reduces multilingual noise and improves overall quality.



Construction Pipeline

- 3k English-origin QA seeds
- 8 target languages: EN, KO, AR, HE, JA, RU, VI, ZH-CN
- 24k total QA instances



What Causes Cross-Lingual Conflicts?

- Most conflicts stem from entity mismatches, showing that models rely heavily on localized knowledge rather than universal reasoning.
- Performance consistently drops on locale-sensitive questions, indicating that cross-lingual inconsistencies are systematic, not random.

Conflict Type	Subtopics (Categories)	Representative Questions	Conflict Count
Entity Conflict	Music, TV actors, Sports players	Who sang Oh What a Night?, Who played TJ on Head of the Class?, Who is the coach for the Toronto Raptors?	1032
Factual Conflict	Geography, Political history, Team records	How many states does the Rocky Mountains cover?, When was the last time the Lakers made the playoffs?	431
Cultural Reference	TV show winners, Music awards, Famous media	Who won America's Got Talent in 2015?, Who has the most Grammys?	512
Ambiguous Question	Religion, Social media, General trivia	Who wrote the Book of Lamentations?, Who has the most Instagram followers?	381

Main results

- English consistently achieves the highest scores, confirming the dominance of English-centric training in multilingual LLMs, while non-English languages—especially Arabic and Hebrew—show substantial degradation, revealing limited cross-lingual transferability.
- Model performance drops drastically on locale-conflict subsets, where semantically identical questions yield different answers across languages, showing that semantic alignment alone cannot guarantee consistent reasoning.

Lang	Oracle LM		Gemma3 12B		Qwen3 14B		LLaMA3.1 8B		EXAONE 7.8B	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
en	89.11	90.97	43.26	52.68	40.73	49.43	40.38	50.56	31.44	39.64
ar	87.86	90.05	18.54	23.62	11.83	19.30	8.53	16.92	3.98	6.04
he	88.30	90.46	20.05	24.83	11.04	16.08	11.86	16.60	5.52	7.20
ja	88.45	92.50	22.81	45.10	19.74	44.03	9.10	37.73	7.34	26.22
ru	87.83	89.54	28.52	35.20	17.67	27.97	14.53	24.18	7.41	9.91
ko	86.73	88.29	22.18	26.56	15.44	19.91	11.55	15.68	15.81	20.32
zh_cn	89.68	93.41	16.22	37.91	26.39	47.57	11.58	36.48	7.66	25.22
vi	89.39	91.19	36.55	44.77	26.83	39.34	26.45	38.38	10.95	14.70
Avg.	88.42	90.80	26.02	36.33	21.21	32.95	16.75	29.57	11.26	18.65

Lang	GEMMA3 12B				QWEN3 14B				EXAONE 7.8B			
	Non-Conflict		Least-Conflict		Non-Conflict		Least-Conflict		Non-Conflict		Least-Conflict	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
en	59.09	72.25	37.69	45.79	64.02	75.36	32.51	40.28	53.67	65.28	23.60	30.59
ar	37.06	46.26	12.01	15.64	27.80	40.54	6.20	11.80	8.90	12.21	2.25	3.86
he	38.63	47.18	13.50	16.94	23.71	32.16	6.58	10.41	9.63	12.49	4.07	5.33
ja	41.16	67.03	16.34	37.36	41.03	65.30	12.22	36.53	15.28	38.24	4.54	21.98
ru	47.41	57.02	21.86	27.50	30.69	49.38	13.07	20.42	10.95	15.48	6.15	7.95
ko	39.35	46.06	16.13	19.69	31.05	38.11	9.93	13.49	30.93	38.48	10.48	13.91
zh_cn	27.32	56.12	12.31	31.49	50.54	71.87	17.87	39.00	15.16	37.15	5.01	21.01
vi	51.62	63.79	31.24	38.07	41.40	61.34	21.69	31.58	18.05	24.30	8.45	11.31
Average	42.70	56.96	20.13	29.06	38.78	54.26	15.01	25.44	20.32	30.45	8.07	14.49